

Risky Business: Quantitative Portfolio Management for Optimized Risk

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Abstract

We backtested five portfolio constructions across normal markets, growth periods, and financial crises using daily Yahoo Finance data frozen through June 28, 2026. The project compares cumulative return, volatility, Sharpe ratio, Sortino ratio, recovery behavior, and benchmark performance against SPY and Mkt-RF. Results show that high-growth portfolios can create strong upside but larger drawdown exposure, while defensive strategies may improve stability at the cost of long-run growth. The project provides a reproducible framework for evaluating portfolio resilience, not investment advice.

Introduction & Motivation

Problem: Return alone does not show how much risk a portfolio takes during market stress.

Research Question: How can portfolio optimization help stakeholders preserve return potential while reducing drawdowns during financial crises and uncertainty?

Stakeholders: Risk-averse investors, institutional funds, portfolio managers, finance students, and analysts who need to evaluate performance beyond simple return.

Portfolios: Barbell, Mega-Cap Technology, Energy, Real Estate, and FMCG. Benchmarks are SPY and Mkt-RF.

Full ticker list and weights are available in the repository.

Background & Related Work

Modern portfolio theory argues that diversification can reduce risk by combining assets with different return patterns. However, crisis periods often challenge diversification because correlations between assets may rise during market stress. Risk-adjusted measures such as the Sharpe ratio and Sortino ratio provide a more complete view of performance than return alone because they account for volatility and downside risk.

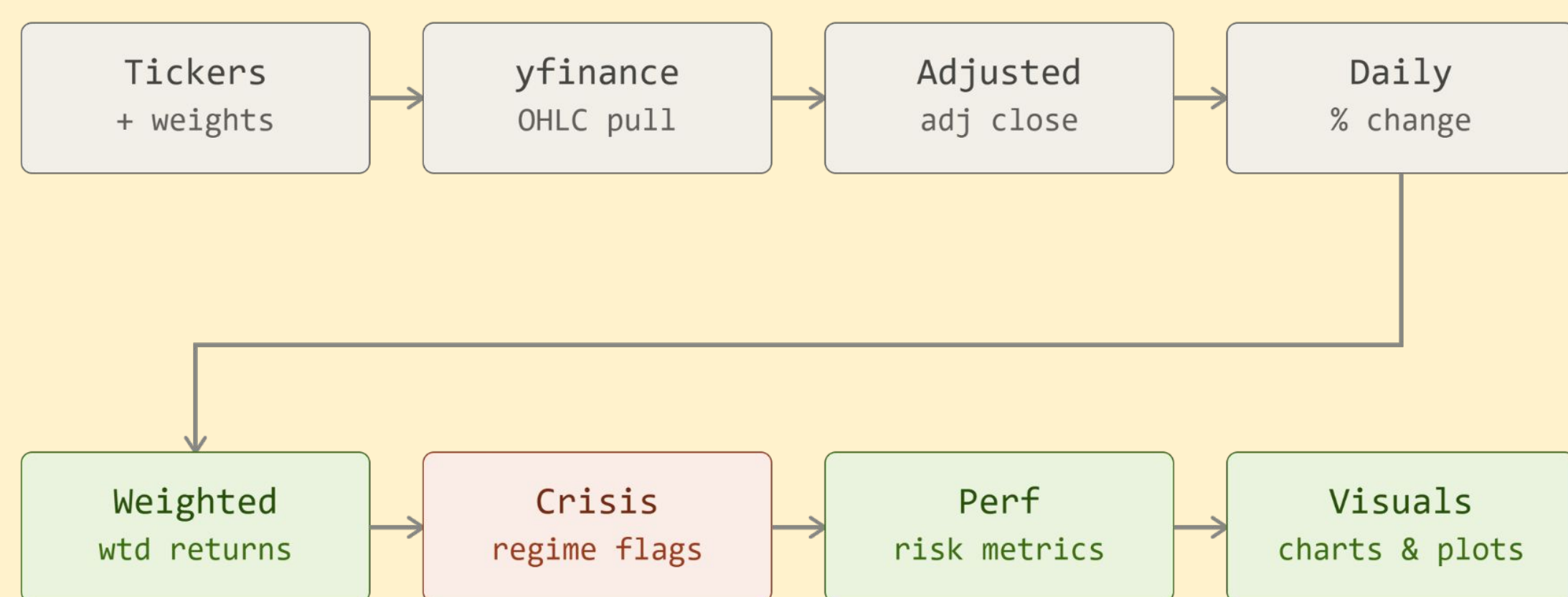
This project builds on portfolio theory, backtesting, and crisis-period analysis to test whether portfolio return behavior becomes less predictable during stressful market periods.

Data, Engineering & Ethics

Sources & scale: The project uses historical daily market data from Yahoo Finance accessed through the `yfinance` Python package. The M2 data scope is frozen through **2026-06-28**. No additional market data will be collected after this milestone.

Ethics & responsible use: All data used in this project are public financial market data. The project does not use personally identifiable information, private investor records, or restricted financial data.

Pipeline: Ticker list + portfolio weights → `yfinance` price pull → adjusted prices → daily returns → weighted portfolio returns → crisis labels → performance metrics → visualizations



Methods & Evaluation

Approach: This project combines backtesting, risk analysis, and barbell strategy. Barbell strategy involves putting a large majority of a portfolio in safe assets such as treasury bills, and a small percentage in options that make large gains in times of high volatility.

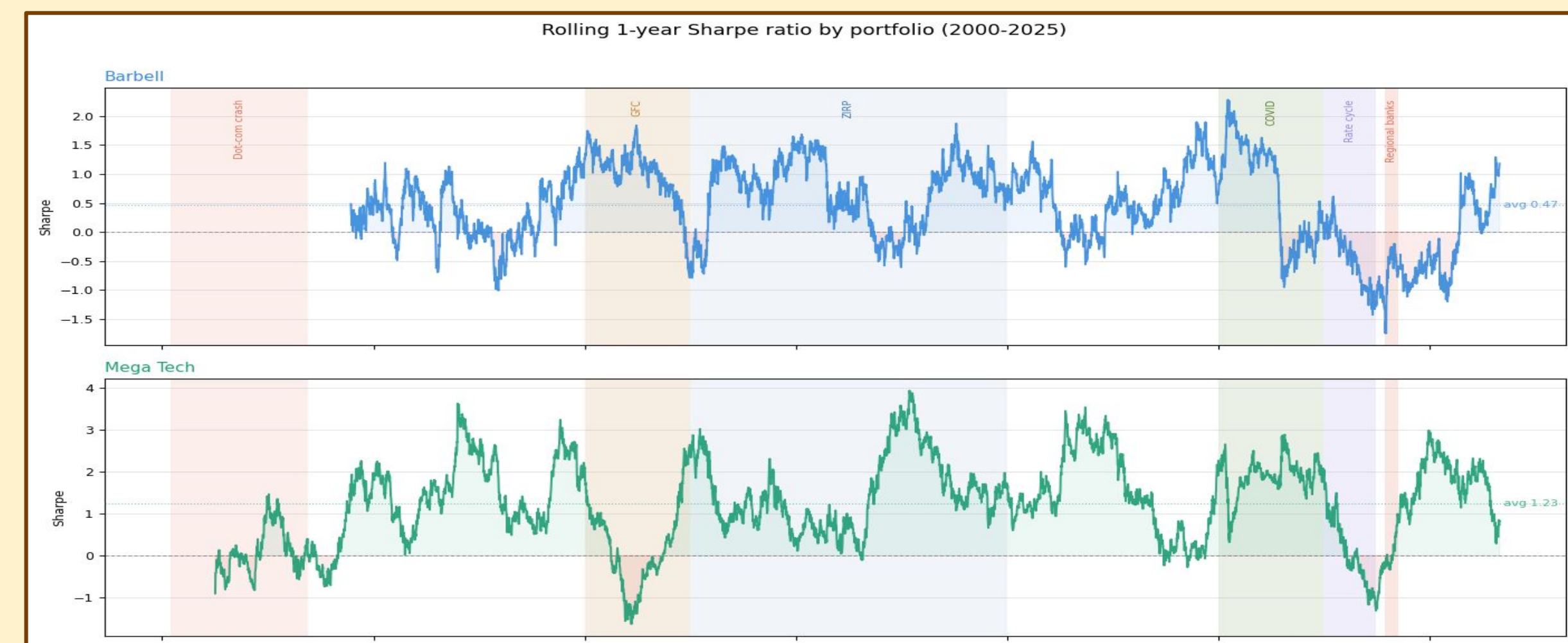
Validation: Each portfolio is evaluated over the same historical period using daily returns. Portfolio returns are calculated by applying balanced portfolio weights to individual asset returns.

Risk and Performance Metrics:

The project evaluates each portfolio using:

- Cumulative return
- Annualized volatility
- Sharpe ratio - Standard measure of return/risk. Quantified as Avg. Return/StDev
- Sortino ratio - The Sharpe Ratio, but only quantifying negative risk
- Recovery behavior
- Benchmark comparison

Results & Visualizations



Barbell				
Regime	Sharpe	Sortino	Ann.Ret	Ann.Vol
Dot-com crash	3.720	6.908	45.70%	11.85%
Recovery	0.406	0.664	6.11%	8.07%
GFC	0.577	1.057	7.88%	12.38%
Recovery/ZIRP	0.817	1.584	8.58%	10.42%
Pre-COVID	0.601	1.167	7.00%	9.49%
COVID	0.802	1.552	8.90%	10.86%
Rate cycle	-1.259	-2.368	-11.15%	19.44%
Regional banks	0.482	0.930	11.43%	13.64%
Post-crisis	0.287	0.507	7.72%	9.76%

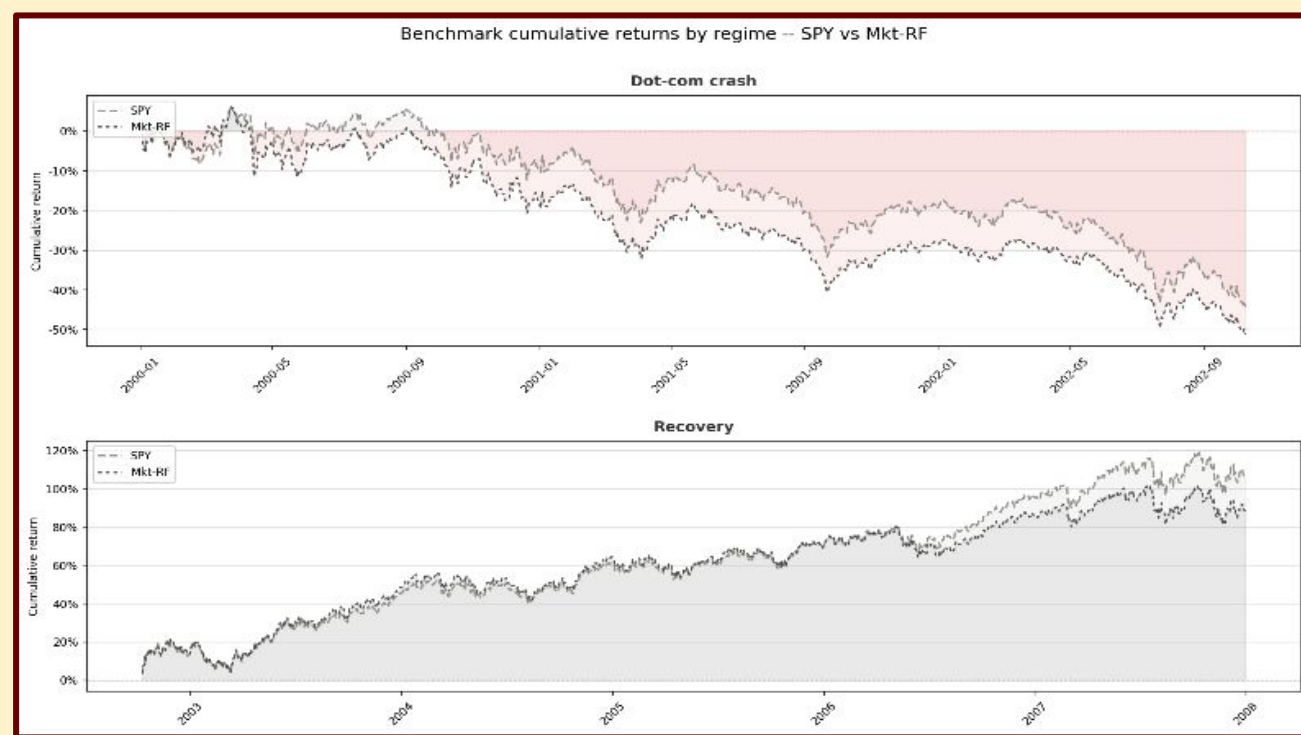


Figure 1. The rolling sharpe ratio by portfolio for available data between 2000-2025
Figure 2 and 3. Cumulative returns for each drawdown and recovery regime.
Figure 4. Sharpe, Sortino, Annual Return, and Annual Volume for each portfolio
Figure 5. Benchmark performance, S&P 500 and Market minus Risk-Free rate (Mkt - Rf).

Discussion, Limitations & Conclusions

Discussion: The project evaluates portfolio resilience from multiple angles rather than relying on return alone. A tech-based portfolio may generate high cumulative returns but still expose stakeholders to large drawdowns during crises. Similarly, a defensive portfolio may reduce downside risk but underperform during growth periods.

Limitations & future work: Historical performance does not guarantee future performance.

- Backtesting can create false confidence if results are overinterpreted.
- Selected assets may introduce survivorship bias.
- Some assets have shorter histories due to IPO dates or fund creation dates.
- Yahoo Finance data may revise adjusted prices or change ticker availability.

Conclusions: Portfolio resilience should be evaluated using multiple dimensions of performance, including return, volatility, drawdown, risk-adjusted return, and forecast stability. Careful diversification across sector and strategy can help give stable returns and help safeguard against volatility.

Reproducibility & References

Repository: [<https://github.com/awjackson00/data510--Quant-Back-Testing-1>] · Key dependencies: [numpy, pandas, tidyfinance, seaborn, scikit-learn, yfinance, statsmodel]

References: Markowitz, H. *Portfolio Selection*. Sharpe, W. *Capital Asset Prices and Risk-Adjusted Return*. Sortino, F. *Downside Risk Framework*. Box and Jenkins/ARIMA time-series modeling. Taleb, N. *Antifragile: Things That Gain From Disorder*. Yahoo Finance and TidyFinance Documentation
"This is not financial advice. Just the work of two grad students who spend way too much time tracking markets."

